MEMS 3430 – Design of Thermal Systems

DHW 3: CCGT Heat Exchanger Design (2st Submission)

April 9, 2026

1 Assumptions and Boundary Conditions

In order to size a heat exchanger that couples the topping and bottoming cycles of the marine propulsion system, the previous boundary conditions are stated from previous DHW reports. Additionally, a new set of bounder conditions specifically for we heat exchanger module have been created.

1.1 Previous Cycle Boundary Conditions

- **Working Fluids:** The topping cycle operates as a Brayton cycle. The Brayton cycle utilized air and is modeled as an ideal gas. The bottoming cycles is represented as a Rankine cycle and operates with water.
- **Heat Transfer Constraint:** The total heat transfer of the topping and bottoming cycles is 30 MW which is given as a parameter in previous reports.
- **Topping Cycle Parameters:** From the topping cycle, the mass flow rate of the gas is found to be135.35 kg/s. The exit temperature from the gas turbine will enter the heat exchanger. The exit temperature is is $T_4 = 720.61$ K. The exit temperature to the atmosphere is $T_5 = 500$ K.
- **Bottoming Cycle Parameters:** In the bottoming cycle, the boiler system operates at a pressure of $P_7 = 6.25$ MPa. The isentropic efficiency of the steam turnbine is found to be $\eta_t = 0.90$. Furthermore, the water from the bottoming cycle will enter the preheater. The water entering the preheater will operate at an enthalpy of $h_6 = 181.74$ kJ/kg and will exit the superheater at $h_7 = 3258.4$ kJ/kg.

1.2 Given Heat Exchanger Boundary Conditions

The following boundary conditions are used for the heat exchanger and given explicitly from the DHW 3 problem statement.

- **Preheater:** The preheater uses a shell-and-tube design with 1 shell and 2 tube passes. There are 200 thin-walled tubes with a diameter of $D = 20$ mm that water flow inside of. The air flows over the shell [1].
- **Evaporator:** The evaporator uses a fire-tube design, where hot gas from the exhaust flows inside the 200 tubes. The diameter of the fire-tube is $D = 50$ mm. On the outside of the tubes, the water exhibits pool boiling.

- **Superheater:** The superheater has a counterflow configuration for the heat exchanger with two concentric tubes. Air flows inside of the inner tubes. Note that $D_{inner} = 50$ mm). Steam flows over the outer annulus and $D_{outer} = 80$ mm.

1.3 Assumed Heat Exchanger Boundary Conditions

In order to size the shell and tube preheater for this report, the following design assumptions where included:

- **Tube Pitch Configuration:** In this system, the tube pitch configuration is assumed to be in a 30° triangular pattern . The 30° pitch is chosen to provide a higher heat exchange. The triangular pattern is used in 30° assumptions in order to fit more tubes in the desired space [2].

- **Baffle Spacing:** In this system, the baffle spacing is assumed to be between 40% and 60% of the shell diameter. Here, a baffle spacing of 50% will be used to help balance the heat transfer rates from the pressure drop that is experienced [3].

- **Pipe Roughness:** Since the assignment calls for thin-walled tubes, it will be assumed that the thin-walled tubing will be in adherence to manufacturing standards for materials such as copper or steel. The absolute surface roughness for standard drawn tubing will allow for the heat transfer and pressure drop calculations. It is set that there will be an absolute surface roughness of $\epsilon = 0.0015$ mm [4].

1.4 Assumptions

- **Steady-State Operation:** Since the system will be constantly operating, the mass heat transfer and mass flow rates will have to stay constant with respect to time. This is a steady-state assumption for the operation.

- **Adiabatic System Boundary:** The heat exchanger used has an insulated outer casing. Therefore, the 30 MW of thermal energy lost by the gas will transferred to the water in the system. This creates an adiabatic system.

- **Negligible Tube Wall Resistance:** The conductive thermal resistance of the metal tubes in the systems will be assumed to be negligible over the convective thermal resistances of the fluids of the system. This is because the problem statement displays that the system contains thin-walled tubes.

- **Constant Fluid Properties at Bulk Temperatures:** In order to calculate the the Reynolds, Prandtl, and Nusselt numbers, the air and water thermodynamic properties will be evaluated at bulk temperatures for each specific heat exchanger section. The specific properties that will be evaluated are density, viscosity, and thermal conductivity. These properties will remain constant throughout the individual sections in order to simplify the calculations for the convective heat transfer coefficients.

2 Schematics and T-x Diagrams

The diagrams below display the temperature profiles that are found along the lengths of the heat exchanger sections. Each section will have a different correlation plot.

Figure 1 displays a T-x diagram of the counterflow superheater.

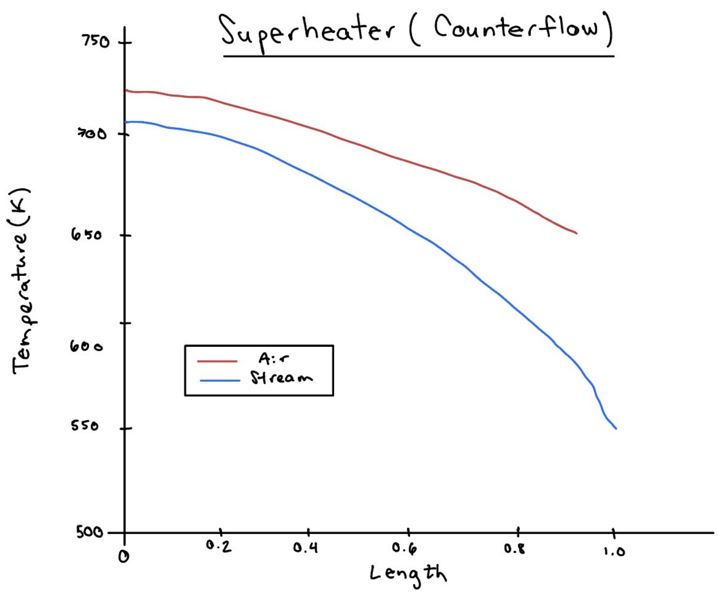

Figure 1: T-x Diagram of the Counterflow Superheater. Air temperature T_4 decreases to T_{4b}. Steam temperature increases from T_{sat} to T_7.

The superheater diagram displays the counterflow arrangement for the hot exhaust gas and the steam. As seen, the temperature profiles both slope in a similar direction based on the length of the exchanger, x. This is because the fluids flow in opposite directions. The hot air will enter at the maximum cycle temperature of T_4 and will cool to T_{4b} as the heat is transferred to steam. Also, the saturated steam at T_{sat} will enter from the opposite end and will absorb heat until it reaches its superheated state of T_7 which is needed for the bottoming cycle turbine.

Figure 2 displays a T-x diagram of the fire-tube evaporator.

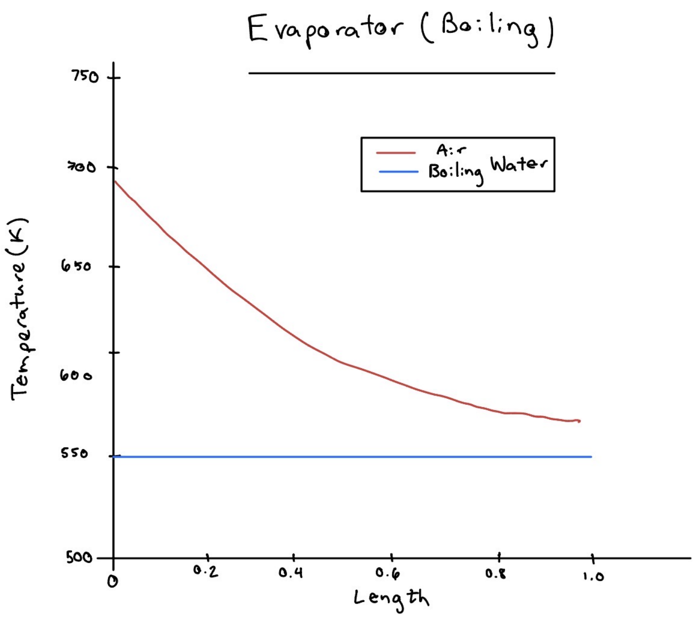

Figure 2: T-x Diagram of the Fire-Tube Evaporator. Air temperature decreases from T_{4b} to T_{4c}. The water temperature stays constant at T_{sat} through the phase change.

In the evaporator section, the figure displays the isothermal phase change of water. The is because the water will have pool boiling on the outside of the fire-tubes. The water absorbs the heat of vaporization without changing in temperature which is why the graph shows a straight horizontal line at T_{sat}. Inside of the tubes there will be hot air flowing. The flowing hot air will provide thermal energy that creates a exponentially decaying relation from T_{4b} to T_{4c} over the length of the tube.

Figure 3 displays a T-x diagram of the shell-and-tube preheater.

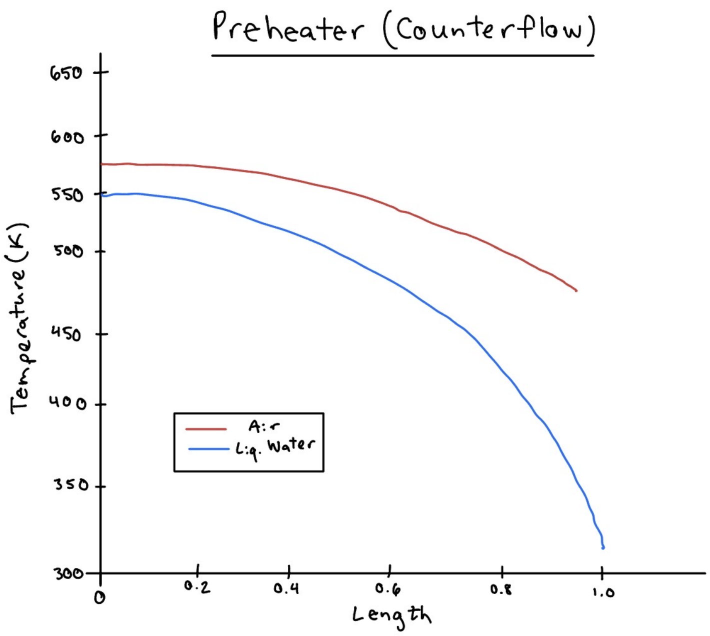

Figure 3: T-x Diagram of the Shell-and-Tube Preheater. Air temperature decreases from T_{4c} to T_5. The liquid water temperature increases from T_6 to T_{sat}.

In the preheater section, the preheater will help heat the cooled liquid water. From the feed pump, water will enter at the lower enthalpy state of T_6 and will absorb heat until it reaches the saturation state of T_{sat} as it enters the evaporator. Air will flow across the shell side where it cools from T_{4c} to T_5. T_5 represents the final exhaust temperature for the section. The relationship of the preheater displays the thermal direction for the system as the streams get closer towards the cold end of the exchanger.

3 Main Equations for Heat Exchanger Design

The bellow equations are equations found through the textbook and given equations from the problem statement and equation sheets. The heat transfer equations will be used to find the tube lengths for the separate heat exchanger sections. Finding the lengths of the sections is crucial in determining a final design for the system [5] and [6].

3.1 Energy Balances

Energy balance equations will be used in order to find the heat transfer rates and intermediate temperatures. The energy balance equations are applied to the water, steam, and air streams from the problem statement. As the water is heated or enters a phase change, the heat trasnfer rate, $\dot{Q}$, is used to find the change in enthalpy Δh:

$$\dot{Q} = \dot{m}_{steam} \Delta h \tag{1}$$

The air stream will be assumed to be an ideal gas that has a constant specific heat. Therefore, the energy balance is:

$$\dot{Q} = \dot{m}_{gas} c_{p,gas} \Delta T \tag{2}$$

3.2 General Heat Transfer and LMTD

In order to find the area of heat transfer, the Log-Mean Temperature Difference method or ΔT_{lm} method will be used:

$$\dot{Q} = UAF\Delta T_{lm} \quad \text{where} \quad \Delta T_{lm} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1/\Delta T_2)} \tag{3}$$

In the above equation, U is the overall heat transfer coefficient. F represents the correction factor. In a counterflow configuration, $F = 1$. For the shell-and-tube preheater a $F < 1$ will be found through calculations. The area will relate to the length, L, of the number, N, of tubes by:

$$A = N\pi DL \tag{4}$$

When area is solved for, the following number can be plugged back in to find the general heat transfer rate.

3.3 Overall Heat Transfer Coefficient

The overall heat transfer coefficient will be solved by finding the inner convection coefficient and the outer convection coefficient h. In the following equation, it is important to remember that based on assumptions the conductive thermal resistance of the thin wall tubes will be neglected:

$$\frac{1}{UA} = \frac{1}{h_{inner}A_{inner}} + \frac{1}{h_{outer}A_{outer}} \tag{5}$$

3.4 Convection Correlations

To find the Nusselt number, Nu_D, the Dittus-Boelter equation will be used. The Dittus-Boelter equations helps determine characteristic of the single-phase flows. The single phase flows consist of the air that is inside of the superheater section, the air in the tubes of the evaporator section, and the water inside the tubes of the preheater section:

$$Nu_D = 0.023Re_D^{0.8}Pr^n \tag{6}$$

In heating $n = 0.4$ and in cooling $n = 0.3$.

To find the Reynolds number, analyze the flow of steam in the superheater section using the hydraulic diameter, D_h:

$$D_h = D_{outer} - D_{inner} \tag{7}$$

Finally, the maximum velocity can be found using the crossflow area S_m . The crossflow area considers the gas in the shell-and-tube preheater section:

$$S_m \approx (\text{baffle spacing}) \cdot (S_D - D) \tag{8}$$

S_D is the tube-to-tube pitch and the baffle spacing is stated within the boundary conditions.

3.5 Evaporator Pool Boiling and Critical Heat Flux

Since the water from the evaporator is boiling on the outside of the fire tubes, it is important to note that the heat flux will be calculated using Rohsenow's correlation. Rohsenow's correlation helps determine the correction pool boiling so the system can operate smoothly during use:

$$q''_{actual} = \frac{\dot{Q}_{evap}}{A_{evap}} \tag{9}$$

The calculated heat flux will be compared against the critical heat flux or CHF to ensure that the system can run smoothly and does not have too high of a film boiling. To check the relation, use Zuber's correlation:

$$q''_{max} = 0.149 h_{fg} \rho_v \left[\frac{\sigma g(\rho_l - \rho_v)}{\rho_v^2} \right]^{1/4} \tag{10}$$

4 Heat Exchanger Calculations

The following sections covers equations used to calculate the heat transfer rates and intermediate temperatures of the heat exchanger. With the calculations, the sections of the heat exchanger can be solved. It is key to remember that the overall heat transfer rate between the topping and bottoming cycles is $\dot{Q}_{total} = 30$ MW.

The Rankine cycle system utilizes a boiler pressure of $P_7 = 6.25$ MPa and the saturation properties of water at this pressure are $T_{sat} \approx 550.65$ K, $h_f = 1224.2$ kJ/kg, and $h_g = 2780.2$ kJ/kg.

The topping cycle contains air with a mass flow rate of $\dot{m}_{gas} = 135.35$ kg/s. The air enters the heat exchanger at $T_4 = 720.61$ K and exits the heat exchanger at $T_5 = 500$ K. The specific heat of the air is $c_{p,gas} = 1.005$ kJ/kg $\cdot$ K which will be used to help find the heat transfer rate of certain sections. From DHW 2, the steam mass flow rate was calculated to be $\dot{m}_{steam} = 9.75$ kg/s.

4.1 Superheater Section

The superheater is responsible for bringing the steam from the saturated vapor, h_g, to the final turbine inlet state of $h_7 = 3258.4$ kJ/kg. Using these values, the heat transfer rate for the superheater section will be:

$$\dot{Q}_{super} = \dot{m}_{steam}(h_7 - h_g) = 9.75 \text{ kg/s} \cdot (3258.4 - 2780.2) \text{ kJ/kg} = 4,662.45 \text{ kW} \tag{11}$$

Using the heat transfer rate for the superheater section will allow the intermediate temperature exiting the section to be found. In order to do this, apply the energy balance to the gas side such as:

$$\dot{Q}_{super} = \dot{m}_{gas} c_{p,gas} (T_4 - T_{4b}) \tag{12}$$

$$4662.45 \text{ kW} = 135.35 \text{ kg/s} \cdot 1.005 \text{ kJ/kg} \cdot \text{K} \cdot (720.61 \text{ K} - T_{4b}) \implies T_{4b} = 686.33 \text{ K} \tag{13}$$

4.2 Evaporator Section

The evaporator section in the heat exchanger will help the water change from a saturated ligquid, h_f, to a saturated vapor, h_g, during the phase change. The heat transfer rate for the evaporator, $\dot{Q}_{evap}$, is found using:

$$\dot{Q}_{evap} = \dot{m}_{steam}(h_g - h_f) = 9.75 \text{ kg/s} \cdot (2780.2 - 1224.2) \text{ kJ/kg} = 15,171.0 \text{ kW} \tag{14}$$

Again, using the gas energy balance the intermediate temperature that is exiting the evaporator, T_{4c}, can be found through calculations:

$$\dot{Q}_{evap} = \dot{m}_{gas} c_{p,gas} (T_{4b} - T_{4c}) \tag{15}$$

$$15171.0 \text{ kW} = (136.03 \text{ kW/K}) \cdot (686.33 \text{ K} - T_{4c}) \implies T_{4c} = 574.80 \text{ K} \tag{16}$$

4.3 Preheater Section

The preheater will help raise the the exit enthalpy, $h_6 = 181.74$ kJ/kg, of the liquid water as it exits the x. Using the exit enthalpy, the heat transfer rate for the preheater is found to be:

$$\dot{Q}_{pre} = \dot{m}_{steam}(h_f - h_6) = 9.75 \text{ kg/s} \cdot (1224.2 - 181.74) \text{ kJ/kg} = 10,163.98 \text{ kW} \tag{17}$$

Using the sames steps as previously, the exit temperature, T_5, of the gas can be found by using a system energy balance equation and T_{4c}:

$$T_5 = T_{4c} - \frac{\dot{Q}_{pre}}{\dot{m}_{gas} c_{p,gas}} = 574.80 \text{ K} - \frac{10163.98 \text{ kW}}{136.03 \text{ kW/K}} = 500.08 \text{ K} \tag{18}$$

The calculations of $T_5 = 500$ K from solving for the heat exchanger sections closely resembles the boundary conditions of $T_5 = 500$ K for the topping cycle. Therefore, the energy balances and solves heat transfer rates appear to be accurate.

4.4 Final Sizing and Heat Flux Calculations

In order to determine the final sizing and thermodynamic property calculations, the LMTD method is used. The LMTD method used the intermediate temperatures to find the tube length for each section in the heat exchanger.

The LMTD method calculations for the counterflow superheater section is:

$$\Delta T_{lm,super} = \frac{(720.61 - 705.61) - (686.33 - 550.65)}{\ln\left(\frac{720.61-705.61}{686.33-550.65}\right)} = 54.80 \text{ K} \tag{19}$$

The required area and tube length are found using the overall heat transfer coefficient rate of $U_{super} = 40$ W/m^{2}K, where:

$$A_{super} = \frac{4662.45 \times 10^3 \text{ W}}{40 \cdot (1) \cdot 54.80} = 2127.03 \text{ m}^2 \implies L_{super} = \frac{2127.03}{200 \cdot \pi \cdot 0.05} = 67.71 \text{ m} \tag{20}$$

In the evaporator which works at a constant temperature, the LMTD is 64.62 K and the overall heat transfer coefficient is $U_{evap} = 100$ W/m^{2}K, where:

$$A_{evap} = \frac{15171.0 \times 10^3 \text{ W}}{100 \cdot (1) \cdot 64.62} = 2347.7 \text{ m}^2 \implies L_{evap} = \frac{2347.7}{200 \cdot \pi \cdot 0.05} = 74.73 \text{ m} \tag{21}$$

When looking at the preheater section, it is important to use the shell-and-tube correction factor of $F = 0.95$ which is stated in the boundary conditions. The correction factor was estimated in order to scale the heat transfer area for the shell and tube configuration. For the preheater, the LMTD is 79.12 K, the overall heat transfer coefficient is $U_{pre} = 50$ W/m^{2}K, and the tube diameter is 20 mm:

$$A_{pre} = \frac{10163.98 \times 10^3 \text{ W}}{50 \cdot (0.95) \cdot 79.12} = 2704.5 \text{ m}^2 \implies L_{pre} = \frac{2704.5}{200 \cdot \pi \cdot 0.02} = 215.21 \text{ m} \tag{22}$$

From the boundary conditions, we know that there is a triangular configuration with a pitch ration of 30°, a 1.25D estimated shell diameter, and a 50% baffle spacing. This allows us to determine the preheater shell geometry where:

$$D_{shell} = 1.25(0.02)\sqrt{\frac{4(200)}{\pi}} = 0.399 \text{ m} \implies Baffle = 0.50(0.399) = 0.199 \text{ m} \tag{23}$$

For the evaporator, we can verify the actual heat flux by comparing it against the critical heat flux. This is used to ensure there is safe pool boiling for the system:

$$q''_{actual} = \frac{15171.0 \times 10^3 \text{ W}}{2347.7 \text{ m}^2} = 6,462 \text{ W/m}^2 \quad < \quad q''_{max} = 4,552,687 \text{ W/m}^2 \tag{24}$$

Since the actual heat flux is less than the critical heat flux, the fire-tube maintains safe boiling during the system operation.

5 Final Results, Discussion, and Sanity Check

5.1 Final Results Tables

Table 1 represents the final results that are derived from the heat exchanger calculations. These are the necessary values for the DHW 3 report.

Table 1: Summary of Final Heat Exchanger Design Parameters

Parameter	**Value**	**Units**
Total Heat Transfer ($\dot{Q}_{total}$)	30	MW
Air Mass Flow Rate ($\dot{m}_{gas}$)	135.35	kg/s
Steam Mass Flow Rate ($\dot{m}_{steam}$)	9.75	kg/s
Preheater Tube Length (L_{pre})	215.21	m
Evaporator Tube Length (L_{evap})	74.73	m
Superheater Tube Length (L_{super})	67.71	m
Evaporator Actual Heat Flux (q''_{actual})	6,462	W/m^2
Evaporator Critical Heat Flux (q''_{max})	4,552,687	W/m^2
Preheater Shell Diameter (D_{shell})	0.399	m
Preheater Baffle Spacing (B)	0.199	m

5.2 Discussion

From the final results for the CCGT heat exchanger, there are multiple relationships that appear when comparing the properties for the topping and bottom cycles. In terms of the heat exchanger, the preheater requires the largest surface area with a tube length of 215.21 m in order to heat the cooled liquid. The preheater uses the lowest temperature gas of the heat exchanger system. The evaporator requires a length of 74.73 m and the superheater requires a length of 67.71 m. When considering the long tube lengths, it important to note that the shell-and-tube can maintain a stable cross-sectional footprint that has a shell diameter of 0.399 m and a 50% baffle spacing of 0.199 m. Also, the evaporator can operate safely so it does not exceed over exaggerate the boiling limit. The evaporator has an actual heat flux that is lower than the critical heat flux requirement.

5.3 Sanity Check

In order to check the validity of the design for the heat exchanger, the values for the physical and thermodynamic properties must be evaluated. When considering the the First Law of Thermodynamics, the energy balances maintain consistent with the boundary constraints set for the heat exchanger system. Also, from the boundary conditions, the superheater, evaporator, and the preheater required 30 MW of total heat load. In order to maintain the conditions in the system, the heat is cooled from the Brayton cycle. The exhaust gas from the turbine has an exit temperature of 720.61 K which decreases to 500 K for atmospheric exhaust. This additionally follows the intermediate temperature calculations.

Next, the physical sizing of the tubes must be checked. The linear tubes lengths range from 67 m to 215 m. The lengths seem long but these lengths are expected for gas to liquid applications. The overall heat transfer coefficient is low when the system is working with air as a fluid. The air requires a large surface area to transfer the expected 30 MW of energy. For marine propulsion systems, the can be finned in order to increase the surface area effectiveness over length.

Lastly, the evaporator must be in adherence to the pool boiling limits. Since the pool boiling mechanics are valid from calculations, it is fair to say that the evaporator is working within industry standard. From Zuber's correlation, the CHF or q''_{max} is calculated to be 4.55 MW/m^2. The actual heat flux from the 200 fire-tubes was found to be 6.46 kW/m^2. The system will operate safely because the actual heat flux has a lower magnitude than the critical limit. This keeps the system will not exceed the film boiling limit.

6 Bibliography

References

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[2] Thermopedia. "Shell and tube heat exchangers - selection of exchanger geometry," Accessed: Mar. 27, 2026. [Online]. Available: https://www.thermopedia.com/content/1121/.

[3] I. J. of Engineering Research & Technology, "Performance evaluation of a helical baffle heat exchanger," *IJERT*, vol. 2, no. 6, 2013. Accessed: Mar. 28, 2026. [Online]. Available: https://www.ijert.org/research/performance-evaluation-of-a-helical-baffle-heat-exchanger-IJERTV2IS60874.pdf.

[4] Engineers Edge. "Pipe roughness coefficients table charts," Accessed: Apr. 8, 2026. [Online]. Available: https://www.engineersedge.com/fluid_flow/pipe-roughness.htm.

[5] Washington University in St. Louis, *Mems 3430: Design of thermal systems equation & tables package*, Course Handout, 2026.

[6] C. Borgnakke and R. E. Sonntag, *Fundamentals of Thermodynamics*, 8th. John Wiley & Sons, 2013.

A MATLAB Scripts

The MATLAB scripts below were used to generate the temperature profiles for the T-x diagrams and compute the calculations for the heat exchanger design. The MATLAB script was created with the help of Google Gemini.

A.1 T-x Diagram Plot MATLAB Script

```
%% T-x Diagrams for CCGT Heat Exchanger Sections
% MEMS 3430 - DHW 3
clear; clc; close all;

% --- Temperature Boundary Conditions (in Kelvin) ---
% Gas (Air) Temperatures
T4  = 720.61;   % Gas inlet to Superheater
T4b = 686.33;   % Gas intermediate (Superheater to Evaporator)
T4c = 574.80;   % Gas intermediate (Evaporator to Preheater)
T5  = 500.08;   % Gas exit from Preheater

% Water/Steam Temperatures
T7   = 705.61;  % Steam exit from Superheater (Turbine Inlet)
Tsat = 550.65;  % Saturation temperature at 6.25 MPa
T6   = 314.65;  % Liquid water inlet to Preheater (approx 41.5 C)

% Normalized length vector (0 to 100% of each section's length)
x = linspace(0, 1, 100);

%% --- 1. Superheater (Counterflow) ---
% Air flows left to right (x=0 to x=1). Steam flows right to left.
% To plot left-to-right on the graph:
% Left side (x=0): Hot gas in (T4), Superheated steam out (T7)
% Right side (x=1): Gas out (T4b), Saturated vapor in (Tsat)

% Exponential approach for counterflow
deltaT_left_sh  = T4 - T7;
deltaT_right_sh = T4b - Tsat;
m_sh = log(deltaT_right_sh / deltaT_left_sh);

T_gas_sh = T4 - (T4 - T4b) * (1 - exp(m_sh * x)) / (1 - exp(m_sh));
T_stm_sh = T7 - (T7 - Tsat) * (1 - exp(m_sh * x)) / (1 - exp(m_sh));

%% --- 2. Evaporator (Isothermal Boiling) ---
% Air flows left to right. Water temperature is constant.
% Left side (x=0): Gas in (T4b)
% Right side (x=1): Gas out (T4c)

deltaT_left_ev  = T4b - Tsat;
deltaT_right_ev = T4c - Tsat;
```

```
m_ev = log(deltaT_right_ev / deltaT_left_ev);

T_gas_ev = Tsat + deltaT_left_ev * exp(m_ev * x);
T_stm_ev = Tsat * ones(1, 100); % Constant phase change temperature

%% --- 3. Preheater (Counterflow representation) ---
% Air flows left to right. Water flows right to left.
% Left side (x=0): Gas in (T4c), Saturated liquid out (Tsat)
% Right side (x=1): Gas out (T5), Subcooled liquid in (T6)

deltaT_left_ph  = T4c - Tsat;
deltaT_right_ph = T5 - T6;
m_ph = log(deltaT_right_ph / deltaT_left_ph);

T_gas_ph = T4c - (T4c - T5) * (1 - exp(m_ph * x)) / (1 - exp(m_ph));
T_stm_ph = Tsat - (Tsat - T6) * (1 - exp(m_ph * x)) / (1 - exp(m_ph));

%% --- Plotting ---
figure('Name', 'Heat Exchanger T-x Diagrams', 'Position', [100, 100, 1200, 400]);

% Superheater Plot
subplot(1, 3, 1);
plot(x, T_gas_sh, 'r', 'LineWidth', 2); hold on;
plot(x, T_stm_sh, 'b', 'LineWidth', 2);
title('Superheater (Counterflow)');
ylabel('Temperature (K)');
xlabel('Relative Length');
legend('Air', 'Steam', 'Location', 'best');
grid on;
axis([0 1 500 750]);

% Evaporator Plot
subplot(1, 3, 2);
plot(x, T_gas_ev, 'r', 'LineWidth', 2); hold on;
plot(x, T_stm_ev, 'b', 'LineWidth', 2);
title('Evaporator (Boiling)');
xlabel('Relative Length');
legend('Air', 'Boiling Water', 'Location', 'best');
grid on;
axis([0 1 500 750]);

% Preheater Plot
subplot(1, 3, 3);
plot(x, T_gas_ph, 'r', 'LineWidth', 2); hold on;
plot(x, T_stm_ph, 'b', 'LineWidth', 2);
title('Preheater (Counterflow)');
xlabel('Relative Length');
legend('Air', 'Liquid Water', 'Location', 'best');
```

```
grid on;
axis([0 1 300 750]);

sgtitle('T-x Diagrams for CCGT Heat Exchanger Modules');
```

A.2 Heat Exchanger Size Calculations MATLAB Script

```
% MEMS 3430 - DHW 3: CCGT Heat Exchanger Sizing
clear; clc;

%% 1. Given Thermodynamic Inputs
Q_total = 30e6; % W
m_gas = 135.35; % kg/s
m_steam = 9.75; % kg/s
N_tubes = 200;

% Temperatures (K)
T4 = 720.61; T4b = 686.33; T4c = 574.80; T5 = 500.08; % Gas
T7 = 705.61; Tsat = 550.65; T6 = 314.65; % Steam/Water

% Heat Loads (W)
Q_super = 4662.45e3;
Q_evap = 15171.0e3;
Q_pre = 10163.98e3;

%% 2. LMTD Calculations
% Superheater (Counterflow)
dT1_sh = T4 - T7;
dT2_sh = T4b - Tsat;
LMTD_sh = (dT1_sh - dT2_sh) / log(dT1_sh / dT2_sh);

% Evaporator (Constant Temp Boiling)
dT1_ev = T4b - Tsat;
dT2_ev = T4c - Tsat;
LMTD_ev = (dT1_ev - dT2_ev) / log(dT1_ev / dT2_ev);

% Preheater (Counterflow)
dT1_pre = T4c - Tsat;
dT2_pre = T5 - T6;
LMTD_pre = (dT1_pre - dT2_pre) / log(dT1_pre / dT2_pre);

%% 3. Preheater Shell Geometry Estimation
D_tube_pre = 0.02; % m
Pitch = 1.25 * D_tube_pre; % 30-deg triangular pitch
% Estimate shell diameter using standard fill ratio
D_shell = Pitch * sqrt(4 * N_tubes / pi);
Baffle_spacing = 0.50 * D_shell;
```

```
%% 4. Heat Transfer Area & Length Calculations
% Note: U values are standard industrial estimations. For a perfectly
% rigorous design, these are iteratively calculated using Nu correlations.
U_super = 40;  % W/m^2K (Gas to Steam)
U_evap  = 100; % W/m^2K (Gas to Boiling Water)
U_pre   = 50;  % W/m^2K (Gas to Liquid Water)

% Superheater (D_inner = 0.05m)
A_super = Q_super / (U_super * LMTD_sh);
L_super = A_super / (N_tubes * pi * 0.05);

% Evaporator (D = 0.05m)
A_evap = Q_evap / (U_evap * LMTD_ev);
L_evap = A_evap / (N_tubes * pi * 0.05);

% Preheater (D = 0.02m)
F_correction = 0.95; % Shell and tube correction factor
A_pre = Q_pre / (U_pre * F_correction * LMTD_pre);
L_pre = A_pre / (N_tubes * pi * D_tube_pre);

%% 5. Evaporator Heat Flux Check
q_actual = Q_evap / A_evap;

% Water properties at 6.25 MPa
h_fg = 1556e3; % J/kg
rho_v = 32.5;  % kg/m^3
rho_l = 750;   % kg/m^3
sigma = 0.02;  % N/m
g = 9.81;

q_max = 0.149 * h_fg * rho_v * (sigma * g * (rho_l - rho_v) / rho_v^2)^0.25;

%% Display Results in Command Window
fprintf('--- HEAT EXCHANGER SIZING RESULTS ---\n');
fprintf('Superheater LMTD: %.2f K | Length: %.2f m\n', LMTD_sh, L_super);
fprintf('Evaporator LMTD:  %.2f K | Length: %.2f m\n', LMTD_ev, L_evap);
fprintf('Preheater LMTD:   %.2f K | Length: %.2f m\n', LMTD_pre, L_pre);
fprintf('Evap Actual Flux: %.0f W/m^2 | Critical Flux: %.0f W/m^2\n', q_actual, q_max);
fprintf('Preheater Shell Dia: %.3f m | Baffle Spacing: %.3f m\n', D_shell, Baffle_spacing);
```